

P1. Dependent

$$C_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + C_3 \begin{pmatrix} -4 \\ 0 \\ -4 \end{pmatrix} = 0$$

$$C_1 + C_2 - 4C_3 = 0 \quad \Rightarrow 2C_2 - 4C_3 = 0 \Rightarrow C_3 = \frac{C_2}{2}$$

$$C_1 - C_2 = 0 \Rightarrow C_1 = C_2$$

$$C_1 + C_2 - 4C_3 = 0$$

So that by letting $C_2 = 2$, we get $C_1 = 2, C_3 = 1$

and

$$2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + \begin{pmatrix} -4 \\ 0 \\ -4 \end{pmatrix} = 0$$

P2. Independent

$$C_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + C_2 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + C_3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$C_1 + C_3 = 0$$

$$C_1 + C_2 = 0$$

$$C_2 + C_3 = 0$$

$$\Rightarrow C_1 = C_2 = C_3 = 0$$

P3. Independent

$$C_1 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + C_3 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = 0$$

$$-C_1 + C_2 + C_3 = 0 \Rightarrow 2C_3 = 0 \Rightarrow C_3 = 0$$

$$C_1 - C_2 + C_3 = 0$$

$$C_1 + C_2 - C_3 = 0 \Rightarrow \begin{cases} C_1 - C_2 = 0 \\ C_1 + C_2 = 0 \end{cases} \Rightarrow C_1 = C_2 = 0$$

(4)

P4. Dependent. The number of vectors is greater than the number of dimensions of $\mathbb{R}^3$ (3).

P5. (a) Independent

$$C_1 \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} + C_2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + C_3 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + C_4 \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$C_1 + C_2 = 0$$

$$C_3 + C_4 = 0$$

$$C_1 + C_3 = 0$$

$$C_2 + C_4 = 0$$

$$\Rightarrow C_1 = C_2 = C_3 = C_4 = 0$$

(b) Dependent.

P8. Solution.

i) $y \equiv 0$ is a solution of $\frac{d^3 y}{dt^3} + y = 0$ with $y(0) = 0$.

ii) Let $y(t)$ be a solution of the differential equation. Then

$$L[cy](t) = \frac{d^3(cy)}{dt^3} + cy$$

$$= c \frac{d^3 y}{dt^3} + cy$$

$$= c L[y](t) = 0$$

$\Rightarrow cy(t)$ is also a solution.

iii) Let $x(t), y(t)$ be solutions of the differential equation. Then

$$L[x+y](t) = \frac{d^3(x+y)}{dt^3} + (x+y)$$

$$= L[x](t) + L[y](t)$$

$$= 0$$

$\Rightarrow (x+y)(t)$ is also a solution.

Other axioms are automatically true

$r^3 + 1 = 0$ (characteristic equation)

$$r_1 = -1, r_2 = \frac{1}{2} + i\frac{\sqrt{3}}{2}, r_3 = \frac{1}{2} - i\frac{\sqrt{3}}{2}$$

$$\Rightarrow y_1(t) = e^{-t}$$

$$\begin{cases} y_2(t) = e^{\frac{t}{2}} \cos \frac{\sqrt{3}}{2} t \\ y_3(t) = e^{\frac{t}{2}} \sin \frac{\sqrt{3}}{2} t \end{cases} \rightarrow \text{basis since they are linearly independent}$$

P9. i) $P(t) \equiv 0 \in V$. Since $L[0] = 0$

ii) Let $P(t) \in V$.

$$L[CP] = C P(0) + 2(CP)'(0) + 3(CP)''(0)$$

$$= C L[P]$$

$$= 0 \Rightarrow C P(t) \in V$$

iii) Let $P_1(t), P_2(t) \in V$.

$$L[P_1 + P_2] = (P_1 + P_2)(0) + 2(P_1 + P_2)'(0)$$

$$+ 3(P_1 + P_2)''(0)$$

$$= L[P_1] + L[P_2] = 0$$

$$\Rightarrow P_1 + P_2 \in V$$

Other axioms are automatically true.

$$P(t) = a_0 + a_1 t + a_2 t^2$$

$$P(0) = a_0$$

$$P'(0) = a_1 \quad \left\{ \begin{array}{l} \Rightarrow a_0 + 2a_1 + 6a_2 = 0 \\ \Rightarrow a_0 = -2a_1 - 6a_2 \end{array} \right.$$

$$P''(0) = 2a_2$$

$$\begin{aligned} \Rightarrow \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} &= \begin{pmatrix} -2a_1 - 6a_2 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} -2a_1 \\ a_1 \\ 0 \end{pmatrix} + \begin{pmatrix} -6a_2 \\ 0 \\ a_2 \end{pmatrix} \\ &= a_1 \underbrace{\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}} + a_2 \underbrace{\begin{pmatrix} -6 \\ 0 \\ 1 \end{pmatrix}} \end{aligned}$$

So that

$$P_1(t) = -2 + t$$

$$P_2(t) = -6 + t^2$$

are a basis for V .